THE PAPER BEHIND THE GAME

When Specifications Drift: Measuring Semantic Preservation Across LLM Representations.

Malhar Alpesh Patel and Luiza Corpaci · NeurIPS 2026 Workshop on AI for Verifiable Coding (VeriCodeGen)

LUIZA CORPACI
TU Wien · Apart Research · CAISH MARS V
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corpaci.luiza@gmail.com

IN ONE PARAGRAPH

Verification proves that an implementation satisfies a specification. It does not prove that a specification which has passed through a language model still means what the original meant. The paper measures that second gap in a domain where meaning can be certified: 500 identities about magmas, small algebraic structures, drawn from the Equational Theories Project, whose implication graph says exactly which law entails which. Each law is rendered in 18 deterministic representations, nine models are asked to write it back as an explicit equation, and a formal oracle labels every answer **equivalent**, **weaker**, **stronger** or **incomparable** to the source, or **uncertified** when the answer falls outside the oracle's catalogue. Uncertified is reported separately and never counted as drift.

SETUP

Laws	500 magma identities, stratified by order; a 150-law subset for repeats and follow-ups.
Representations	Canonical infix, LaTeX, prefix, postfix · Lean, TPTP, SMT-LIB · Python lambda, Cayley-table checker · bracket words, nested noun phrases, word problem · SSA, JSON AST, indented tree, DOT · integer slots, confusable names.
Models	Nine, in three roster tiers: Mistral Small 3.2, Qwen3-30B-A3B, Phi-4 · Gemini 2.5 Flash, Llama 4 Maverick, Grok 4.20 · DeepSeek V3.2, GPT-5.2, Claude Opus 5.
Calls	81,000 in the main sweep; 153,000 with repeats and three follow-up studies. Temperature zero and seeds where supported.
Oracle	A historical snapshot of the ETP implication graph, checked in both directions. Coverage failures are a separate outcome, not a verdict.

WHY IT MATTERS FOR VERIFIABLE CODING

- **Preservation is a separate obligation.** A certified weaker reconstruction admits objects the original excluded, and a later proof of that reconstruction cannot restore the lost requirement. Checking that the specification survived is a different check from proving code against it.
- **Keep direction and coverage in the interface.** A single pass/fail conflates opposite inclusion errors with out-of-coverage answers. Show intended-to-generated and generated-to-intended evidence, and an explicit uncertified outcome.
- **Identifiers and instructions deserve evaluation.** Preserve a bijection between source and target variables, check co-reference, and treat documentation and preservation instructions as part of the system under test.

READ THE PAPER

Full text, appendices and reviews on OpenReview.

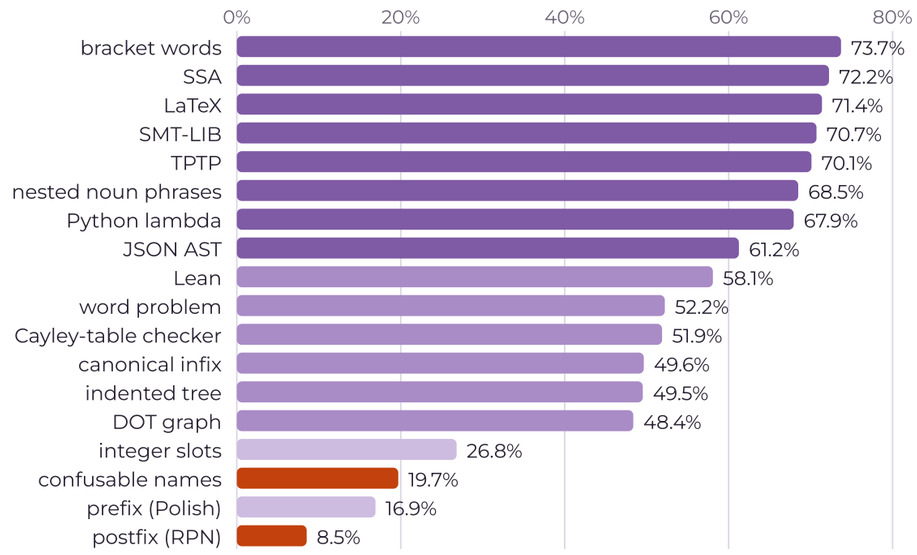
`openreview.net/forum?id=qtgG5HZSsk`

Artifact, data and the certified oracle: **github.com/corpaci/SpecDrift**

WHAT WE FOUND

- **Representation matters enormously.** Certified recovery runs from 73.7% for mechanical bracket-word text to 8.5% for postfix notation, a 65-point span on the same 500 laws. Mechanical English beats symbolic prefix notation with the same word order.
- **Drift has a direction, and it concentrates.** Under confusable identifiers, 1,365 of 4,500 answers (30.3%) are provably weaker than the source; 93.7% of those use fewer distinct variables, consistent with two variables being read as one.
- **Look-alike letters are a cliff.** Across six identifier schemes recovery falls from 64.4% on canonical letters to 3.1% at Latin/Cyrillic homoglyphs; 81% of those answers are weaker, and 87% of merged pairs are a letter and its twin.
- **Reading instructions help, selectively.** Notation notes with examples recover large losses for some model and notation pairs; an explicit anti-transformation instruction improves even a verbatim-input control. Extra context alone does not.
- **Repeated sampling limits fine rankings.** Three samples per item show enough residual variability that single-run model orderings should not be read closely.

CERTIFIED RECOVERY BY REPRESENTATION



Share of 4,500 answers per representation that the oracle certified as equivalent to the source law. Pooled over nine models. Uncertified answers are counted as not recovered, not as wrong.

LIMITS, STATED PLAINLY

Short single equations, with common notation possibly present in pretraining. A fixed historical roster of API models with unequal settings; tier comparisons are descriptive. The oracle is a catalogue snapshot, so uncertified is a coverage statement, not a meaning statement. None of this estimates drift rates in deployed AI systems. What it does establish is that one-step rewriting by capable models produces thousands of certified strict weakenings, and that the direction can be measured.

From the paper to the poster game. The paper measures drift where an oracle exists. The game asks the same question where none does: two published AI specifications, changed one word at a time by visitors and by models, judged by other visitors before and after dictionary help. The paper's vocabulary, same, changed, unresolved, is the game's.

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